

WS Tech – Node.js NPM & Modules Lab Tasks

Lab Task 1: JSON Practice

- 1 Create a file user.json
- 2 Add at least 3 users with name, age, and skills
- 3 Read JSON file in Node.js using require()
- 4 Print user details using console.log()
- 5 Convert JS object to JSON string using JSON.stringify()

Lab Task 2: Initialize NPM Project

- 1 Create a new project folder
- 2 Run npm init -y
- 3 Check package.json file
- 4 Add start script in package.json
- 5 Run project using npm start

Lab Task 3: Install External Package

- 1 Install chalk package using npm install chalk
- 2 Use chalk to print colored messages in console
- 3 Verify dependency added in package.json
- 4 Explain what node_modules folder contains

Lab Task 4: Create Custom Module

- 1 Create math.js file
- 2 Add functions: add, subtract, multiply
- 3 Export all functions using module.exports
- 4 Import in app.js using require()
- 5 Test all functions

Lab Task 5: Build Modular Calculator App

- 1 Create calculator.js module
- 2 Export at least 4 functions (add, sub, mul, div)

- 3 Create app.js to import calculator module
- 4 Take numbers from variables and print results
- 5 Organize project properly

Bonus Task (Apply Level)

- 1 Create data.json file to store product list
- 2 Read file asynchronously using fs.readFile()
- 3 Display product names in console
- 4 Handle error properly

Submission Instructions:

Submit complete project folder with package.json, node_modules (optional), and all .js files. Folder name: WS_Tech_NPM_Modules_StudentName